

Lecture 8a: Time-varying Covariates

March 13, 2025

Some Concepts from Last Lecture...

- ICC
- ANOVA
 - F-test
 - Chisq (Deviance)
- AIC/BIC

ICC

- Intraclass Correlation (ICC): how strongly units in the same group resemble each other, e.g., a type of correlation
- In an intercept-random-effects model:

$$Y_{ij} = \beta_0 + \beta_1 x_{1ij} + \cdots + \beta_p x_{pij} + b_{0i} + \epsilon_{ij}$$

we have

$$Var(Y_{ij}) = Var(b_{0i} + \epsilon_{ij}) = \sigma_b^2 + \sigma^2$$

$$\begin{aligned} Cov(Y_{ij}, Y_{ij'}) &= Cov(b_{0i} + \epsilon_{ij}, b_{0i} + \epsilon_{ij'}) = Cov(b_{0i}, b_{0i}) \\ &= Var(b_{0i}) = \sigma_b^2 \end{aligned}$$

- Thus

$$ICC = corr(Y_{ij}, Y_{ij'}) = \frac{Cov(Y_{ij}, Y_{ij'})}{\sqrt{Var(Y_{ij})Var(Y_{ij'})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}$$

ICC (cont.)

$$ICC = \text{corr} \left(Y_{ij}, Y_{ij'} \right) = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}$$

- ICC= correlation among observations within the class/group
- ICC=ratio of the between cluster/group variance to the total variance
- ICC tells us the proportion of the total variance in Y that is accounted for by the clustering/grouping.

Why ICC is useful?

- It can help determine whether or not a linear mixed model is even necessary. E.g., if $ICC=0$, then it is fine to use a simpler analysis technique, e.g., standard linear regression.
- It can be meaningful to understand how much of the overall variation in the response is explained simply by clustering/grouping. There are two scenarios:
 - **Trait** (varies among people but not within a person on different occasions), or
 - **State** (varies little on average among people, but varies a lot across occasions).
- It can also be meaningful to see how the ICC (and between and within cluster variances) changes as variables are added to the model.

ANOVA

- Consider the problem of testing $H_0: M_0$ (**sub – model**) vs $H_1: M_1$ (**larger model**).
- Assume the models M_0, M_1 have $p_0 < p_1$ parameters
- At the high-level, there are two tests:
 - F-test
 - Chi-square test (deviance/log-likelihood)

ANOVA: General F-test in Linear Models

- More generally, consider the problem of testing $H_0: M_0$ (sub – model) vs. $H_1: M_1$ (larger model).
- Assume the models M_0, M_1 have $p_0 < p_1$ parameters
- Assume $\hat{Y}_0$ and $\hat{Y}_1$ are the fitted values of raw Y under H_0 and H_1 , resp.

- Then the test statistics is

$$F_{obs} = \frac{|\hat{Y}_1 - \hat{Y}_0|^2 / (p_1 - p_0)}{|Y - \hat{Y}_1|^2 / (n - p_1)}$$

and we reject the sub-model M_0 if and only if

- $F_{obs} > F_{p_1 - p_0, n - p_1, \alpha}$

[This is the general form of the **ANalysis Of VAriance** (ANOVA) test]

Loglikelihood

- When the absolute value of the $-2 \times \text{loglike}$ decreases, this is an indication of a better model fit.
- The loglikelihood will always increase (if negative, decrease or get closer to zero) when additional variables are added
 - This is why we examine multiple fit indicators in tandem

Deviance

- In statistics, deviance is a goodness-of-fit statistic for a statistical model (the smaller the better!)

- Deviance for a model M is based on the likelihood functions:

$$\begin{aligned} D(M) &= -2 \log(Y|M) - [-2 \log(Y|M_s)] \\ &= 2 [\log(Y|M_s) - \log(Y|M)] \end{aligned}$$

where M_s is the **saturated** model that fits the data perfectly!

- For the regression context (linear regression or mixed-effects mode)

$$D(M) = \frac{1}{\sigma^2} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 - 0 = \frac{1}{\sigma^2} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$$

Deviance

(Singer and Willett, pgs. 116-122)

(4.15)

$$\text{Deviance} = -2[LL_{\text{current model}} - LL_{\text{saturated model}}]$$

$$\text{Deviance} = -2LL_{\text{current model}}$$

- Deviance statistic compares the loglikelihood of the current and saturated model
- How much worse is the current model than the best model (the saturated model)
- A small deviance means the current model is a close fit with the saturated model. This indicates a nice model fit
- A large deviance means the current model is a poor fit

ANOVA: Chi-Square Test

- Suppose there are two models, $M_0 \subset M_1$ with $p_0 < p_1$ parameters, respectively. Then $D(M_0) \geq D(M_1)$
- When testing $H_0: M_0$ (the smaller model) fits the data as well as M_1 against $H_1: M_1$ fits the data better, the test statistic is

$$\chi_{obs}^2 = D(M_0) - D(M_1) \left[= \frac{1}{\sigma^2} |\hat{Y}_1 - \hat{Y}_0|^2 \right]$$

and reject H_0 if and only if $\chi_{obs}^2 \geq \chi_{p_1-p_0, \alpha}^2$

`[pchisq(chisq.obs, df = p1-p0, lower.tail=FALSE)]`

Deviance

- To compare models....
 1. One must be nested in the other
 2. The same data must be used
 3. The same number of cases must be used in both models
 4. If FML comparing entire models
 5. If REML only comparing the random effects

Deviance Example

- Mean model only
- -2LL = -335.07799

Per the Deviance formula we multiple by 2, which = **670.16**

Mixed-effects ML regression		Number of obs	=	246
Group variable: id		Number of groups	=	82
		Obs per group:		
		min	=	3
		avg	=	3.0
		max	=	3
		Wald chi2(0)	=	.
		Prob > chi2	=	.
Log likelihood = -335.07799				

		Mixed-effects ML regression				Number of obs = 246	
		Group variable: id				Number of groups = 82	
						Obs per group:	
						min = 3	
						avg = 3.0	
						max = 3	
Random-effects Parameters						Wald chi2(1) = 20.11	
id: Identity		Log likelihood = -318.83122				Prob > chi2 = 0.0000	
var(_cons)							
var(Residual)		alcuse	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
LR test vs. linear model: chi2(2) = 78.65		age_14	.2706514	.060358	4.48	0.000	.1523519 .3889509
		_cons	.6513035	.1015525	6.41	0.000	.4522642 .8503428

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Independent					
		var(age_14)	.1172599	.0446745	.0555715 .2474266
		var(_cons)	.5432039	.1170779	.3560423 .8287513
		var(Residual)	.3629466	.0533485	.272099 .4841261
LR test vs. linear model: chi2(2) = 78.65					
Prob > chi2 = 0.0000					

- Growth model
- -2LL = -318.83122
- Per the Deviance formula we multiple by 2, which = **636.61**

Deviance Example

$$670.16 - 636.61 = 33.5$$

33.5 > 16.27 (the .001 critical value for the χ^2 distribution)

We conclude that the unconditional growth model provides a better fit than the mean only model

Mixed-effects ML regression				Number of obs	=	246
Group variable: id				Number of groups	=	82
				Obs per group:		
				min	=	3
				avg	=	3.0
				max	=	3
Log likelihood = -335.07799				Wald chi2(0)	=	.
				Prob > chi2	=	.

Mixed-effects ML regression				Number of obs	=	246
Group variable: id				Number of groups	=	82
				Obs per group:		
				min	=	3
				avg	=	3.0
				max	=	3
Log likelihood = -318.83122				Wald chi2(1)	=	20.11
				Prob > chi2	=	0.0000

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LR test vs. linear model: chi2(2) = 78.65				Prob > chi2 = 0.0000	
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AIC/BIC

- Akaike Information Criterion (AIC)/ Bayesian Information Criterion (BIC)
 - Correct for some of the limitations of singularly using the loglikelihood and the Deviance statistic
- AIC controls for the increase in model parameters
- BIC controls for the number of parameters and sample size
- These models do not need to be nested as long as the same data is utilized
- The smaller the AIC/BIC the better the model

AIC/BIC

- Under MLE, can use both fixed and random effect parameters
- Under REML only random effects
- Art and science
 - Not clear what is a significant drop
 - Should always be used together
 - Conceptual model important

Notes on Model Fit

- Model fit statistics should be taken together. When all are trending in a specific direction, then you have more confidence in your model fit
- There are many more model fit techniques that we will discuss throughout the rest of the course, but these are the basics

Time Varying Covariates

Dr. Cook's lecture slides

What are time-varying covariates?

(Singer and Willet pgs. 159-160)

- Time varying predictors change over time
- These variables may change the value of the outcome at any given measurement occasion
- Examples:
 - How do changes in parental support influence trajectories of depression in the transition to adulthood?
 - How does level of depression influence trajectories of colon cancer mortality in aging adults?

Example for illustrating time-varying predictors: Unemployment & depression

Source: *Liz Ginexi and colleagues (2000), J of Occupational Health Psychology*

■ Sample: *254 people identified at unemployment offices.*

■ Research design: *Goal was to collect 3 waves of data per person at 1, 5 and 11 months of job loss. In reality, however, data set is not time-structured:*

- *Interview 1 was within 1 day and 2 months of job loss*
- *Interview 2 was between 3 and 8 months of job loss*
- *Interview 3 was between 10 and 16 months of job loss*
- *In addition, not everyone completed the 2nd and 3rd interview.*

■ **Time-varying predictor: Unemployment status (UNEMP)**

- *132 remained unemployed at every interview*
- *61 were always working after the 1st interview*
- *41 were still unemployed at the 2nd interview, but working by the 3rd*
- *19 were working at the 2nd interview, but were unemployed again by the 3rd*

■ **Outcome:** *CES-D scale—20 4-pt items (score of 0 to 80)*

■ Research question

- *How does unemployment affect depression symptomatology?*



Main effect of time-varying covariates

(Singer and Willet pg. 161)

- Ginexi et al (2000)
- Examining the influence of unemployment status on depression
- N = 254
- Follow-up at 3-8 or 10-16 months
- CES-D (0-80)
- Unemployment (UNEMP =1, EMPL =0)
 - [1,1,1]; [1,0,0]; [1,1,0]; [1,0,1]
- Time_{ij}: Number of months between the date on initial unemployment for each individual
 - Time_{ij} = 0

Table 5.6: Excerpts from the person-period data set for the unemployment study

ID	MONTHS	CES-D	UNEMP
7589	1.3142	36	1
7589	5.0924	40	1
7589	11.7947	39	1
55697	1.3471	7	1
55697	5.7823	4	1
65641	0.3285	32	1
65641	4.1068	9	0
65641	10.9405	10	0
65441	1.0842	27	1
65441	4.6982	15	1
65441	11.2690	7	0
53782	0.4271	22	1
53782	4.2382	15	0
53782	11.0719	21	1

Main effect of time-varying covariates

(Singer and Willet pg. 163, Table 5.7)

2-level unconditional model

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$

$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \zeta_{1i},$$

(5.3a)
where
(5.3b)

$$\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2) \text{ and } \begin{bmatrix} \zeta_{0i} \\ \zeta_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{10} & \sigma_1^2 \end{bmatrix}\right)$$

Ginexi et al (2000)

		Parameter	Model A
Fixed Effects			
Composite	Intercept	Y ₀₀	17.6694**
model	(initial status)		(0.7756)
	TIME	Y ₁₀	-0.4220***
	(rate of change)		(0.0830)

- The average person has a depression score of 17.67
- On average depression scores decrease at a rate of .42 per month

- All of the variance components are statistically significant
- This suggest we can explore time-varying predictors



Variance Components

Level-1:	Within-person	σ_{ε}^2	68.85***
Level-2:	In intercept	σ_0^2	86.85***
	In rate of Change	σ_1^2	0.36*

Main effect of time-varying covariates

(Singer and Willet pg. 163 and Table 5.7)

Ginexi et al (2000)

Composite for Unconditional Growth Model

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Variance Components			
Level-1:	Within-person	σ^2_{ε}	68.85***
Level-2:	In intercept	σ^2_0	86.85***
	In rate of Change	σ^2_1	0.36*

		Parameter	Model A
Fixed Effects			
Composite	Intercept	γ_{00}	17.6694**
model	(initial status)		(0.7756)
	<i>TIME</i>	γ_{10}	-0.4220***
	(rate of change)		(0.0830)

NOTE: We are going to focus on the composite equation in this class. However, if you are more comfortable with the 2-level equation level see pages 168-169 of Singer and Willet for more information

Main effect of time-varying covariates

(Singer and Willet pg. 164)

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}].$$



- Number of months since job loss (Time)
- Participant i's value of unemployment at different measurement occasions
- Person specific residuals

γ_{10} is the population average monthly rate of change in depression (CES-D) score, controlling for unemployment status

γ_{20} is the population average difference, over time, in depression score between employed and unemployed

γ_{00} The intercept does not make sense to interpret in this case because everyone started out as being unemployed ($UNEMP = 1$). However, the interpretation would be the depression score of someone who is employed on day 1 of job loss

Analytic approach: We're going to sequentially fit 4 increasingly complex models

Goal is to both explain the use of TV predictors and illustrate how you do practical data analysis

Model A: An individual growth model with no substantive predictors

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}, \text{ where } \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$$

Model B: Adding the main effect of UNEMP

$$Y_{ij} = \gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij} + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Model C: Allowing the effect of UNEMP to vary over TIME

$$Y_{ij} = \gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij} + \gamma_{30}UNEMP_{ij} \times TIME_{ij} + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Model D: Also allows the effect of UNEMP to vary over TIME, but does so in a very particular way

$$Y_{ij} = \gamma_{00} + \gamma_{20}UNEMP_{ij} + \gamma_{30}UNEMP_{ij} \times TIME_{ij} + [\zeta_{0i} + \zeta_{2i}UNEMP_{ij} + \zeta_{3i}UNEMP_{ij} \times TIME_{ij} + \varepsilon_{ij}]$$

Main effect of time-varying covariates

(Singer and Willet pg. 165 and pg. 163, Table 5.7)

Plot 1 –Unemployed (1,1,1)

- Predicted change trajectory in CES-D for someone who remains unemployed

Plot 2 –Job 5m (1,0,0)

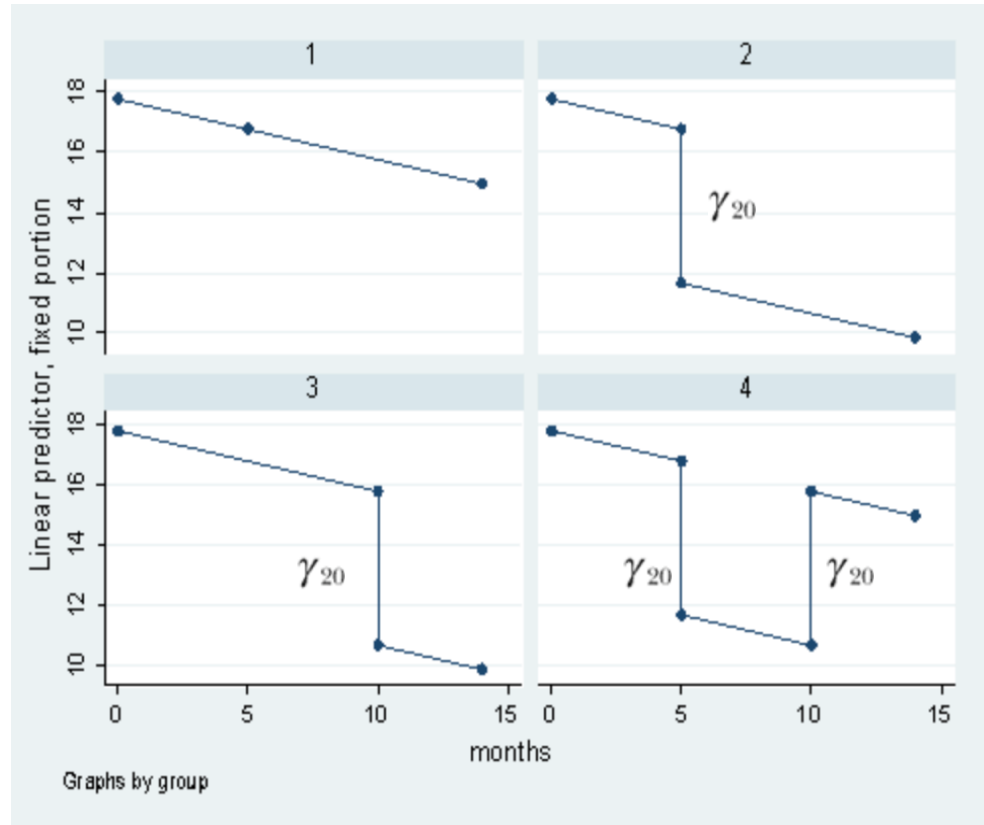
- Hypothesized change from Plot 1 for someone who stayed employed after 5-month assessment

Plot 3 –Job 10m (1,1,0)

- Hypothesized change from Plot 1 for someone who was employed after the 10-month assessment

Plot 4 –Job 5m; loss at 10m (1,0,1)

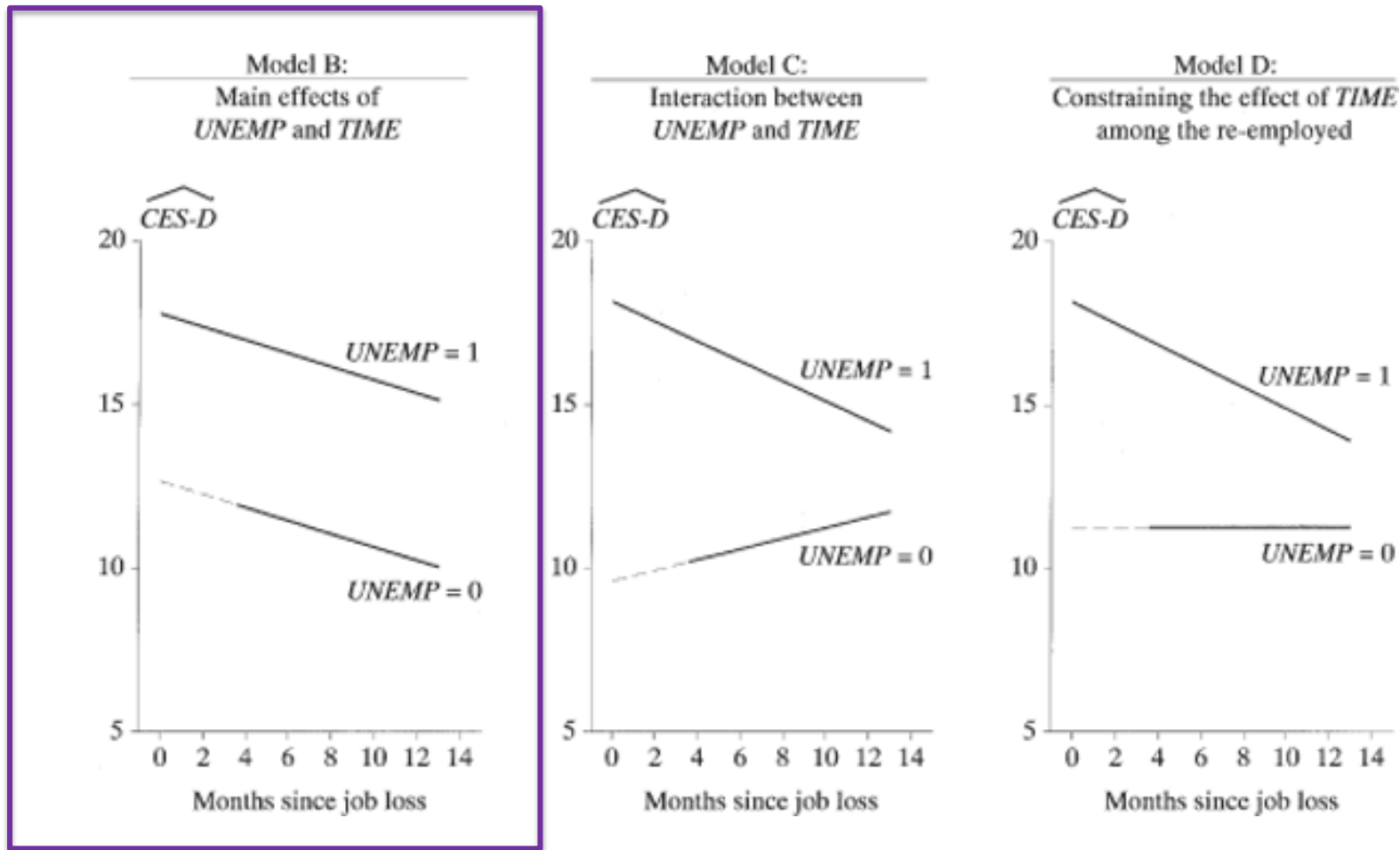
- Hypothesized change from Plot 1 for someone who was employed after the 5-month assessment but again unemployed at the 10-month assessment



NOTE: In this model we have constrained the effect of unemployment to be linear and constant among individuals. Thus, the difference between trajectories will be the same for all individuals. γ_{20} is the difference between the consistently unemployed (1,1,1) and the other patterns of unemployment.

Time-varying predictor and the slope

(Singer and Willet pg. 167, Figure 5.4)



Main effect of time-varying covariates

(Singer and Willet pg. 163, Table 5.7)

Significantly less monthly decline in depression between model A and B (-.422 vs. -.202)

Depression scores, on average, 5.11 higher among the unemployed

Model A: Unconditional growth model

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Model B: Main effect model

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

		Parameter	Model A	Model B
Fixed Effects				
Composite model	Intercept	γ_{00}	17.6694**	12.6656***
	(initial status)		(0.7756)	(1.2421)
	<i>TIME</i>	γ_{10}	-0.4220***	-0.2020*
	(rate of change)		(0.0830)	(0.0933)
		<i>UNEMP</i>	γ_{20}	5.1113***
				(0.9888)
		<i>UNEMP by TIME</i>	γ_{30}	
Variance Components				
Level-1:	Within-person	σ_{ε}^2	68.85***	62.39***
Level-2:	In intercept	σ_0^2	86.85***	93.52***
	In rate of Change	σ_1^2	0.36*	0.46**
	In <i>UNEMP</i>	σ_2^2	—	—
	In <i>UNEMP by TIME</i>	σ_3^2	—	—
Goodness-of-fit				
		Deviance	5133.1	5107.6
		AIC	5145.1	5121.6
		BIC	5166.3	5146.4

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Improved model fit statistics

Fitting and interpreting Model B, which includes the TV predictor UNEMP

	Parameter	Model A	Model B
Fixed Effects			
Composite model	Intercept (initial status)	γ_{00} 17.6694** (0.7756)	12.6656*** (1.2421)
	TIME (rate of change)	γ_{10} -0.4220*** (0.0830)	-0.2020* (0.0933)
	UNEMP		γ_{20} 5.1113*** (0.9888)
	UNEMP by TIME		γ_{30}
Goodness-of-fit			
	Deviance	5133.1	5107.6
	AIC	5145.1	5121.6
	BIC	5166.3	5146.4

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Monthly rate of decline is cut in half by controlling for UNEMP (still sig.)

UNEMP has a large and stat sig effect

Model A is a much poorer fit (Δ Deviance = 25.5, 1 df, $p < .001$)

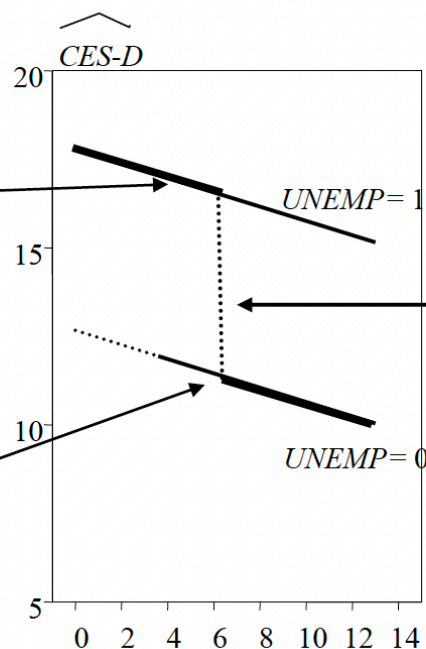
Consistently unemployed (UNEMP=1):

$$\hat{Y}_j = (12.6656 + 5.1113) - 0.2020 \text{MONTHS}_j$$

$$\hat{Y}_j = 17.7769 - 0.2020 \text{MONTHS}_j$$

Consistently employed (UNEMP=0):

$$\hat{Y}_j = 12.6656 - 0.2020 \text{MONTHS}_j$$



What about people who get a job?

What about the variance components?

☛ Variance components behave differently when you're working with TV predictors

When analyzing time-invariant predictors, we know which VCs will change and how:

- ☛ Level-1 VCs will remain relatively stable because time-invariant predictors cannot explain much within-person variation
- ☛ Level-2 VCs will decline if the time-invariant predictors explain some of the between person variation

When analyzing time-varying predictors, all VCs can change, but

- ☛ Although you can interpret a decrease in the magnitude of the Level-1 VCs
- ☛ ☛ Changes in Level-2 VCs may not be meaningful!

		Parameter	Model A	Model B
Fixed Effects				
Composite model	Intercept (initial status)	γ_{00}	17.6694** (0.7756)	12.6656*** (1.2421)
	TIME (rate of change)	γ_{10}	-0.4220*** (0.0830)	-0.2020* (0.0933)
	UNEMP	γ_{20}		5.1113*** (0.0000)

Variance Components				
Level-1:	Within-person	σ^2_ϵ	68.85***	62.39***
Level-2:	In intercept	σ^2_0	86.85***	93.52***
	In rate of change	σ^2_1	0.36*	0.46**

- $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Level-1 VC, σ^2_ϵ

- ☛ Adding UNEMP to the unconditional growth model (A) reduces its magnitude 68.85 to 62.39
- ☛ UNEMP “explains” 9.4% of the variation in CES-D scores

☛ Look what happened to the Level-2 VC's

- ☛ In this example, they've **increased**!
- ☛ Why?: Because including a TV predictor changes the meaning of the individual growth parameters (e.g., the intercept now refers to the value of the outcome when **all** level-1 predictors, including UNEMP are 0).

We can clarify what's happened by decomposing the composite specification back into a Level 1/Level-2 representation

Important Note: Because you can no longer rely on the Level 2 residuals, you must rely on the Level – 1 residuals and the goodness of fit statistics!

Model C: Might the effect of a TV predictor vary over time?

When analyzing the effects of time-invariant predictors, we automatically allowed predictors to affect the trajectory's slope

Because of the way in which we've constructed the models with TV predictors, we've automatically constrained UNEMP to have only a "main effect" influencing just the trajectory's level

To allow the effect of the TV predictor to vary over time, just add its interaction with TIME

$$Y_{ij} = \gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij} + \boxed{\gamma_{30}UNEMP_{ij} \times TIME_{ij}} + \boxed{[\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]}$$

Two possible (equivalent) interpretations:

- *The effect of UNEMP differs across occasions*
- *The rate of change in depression differs by unemployment status*

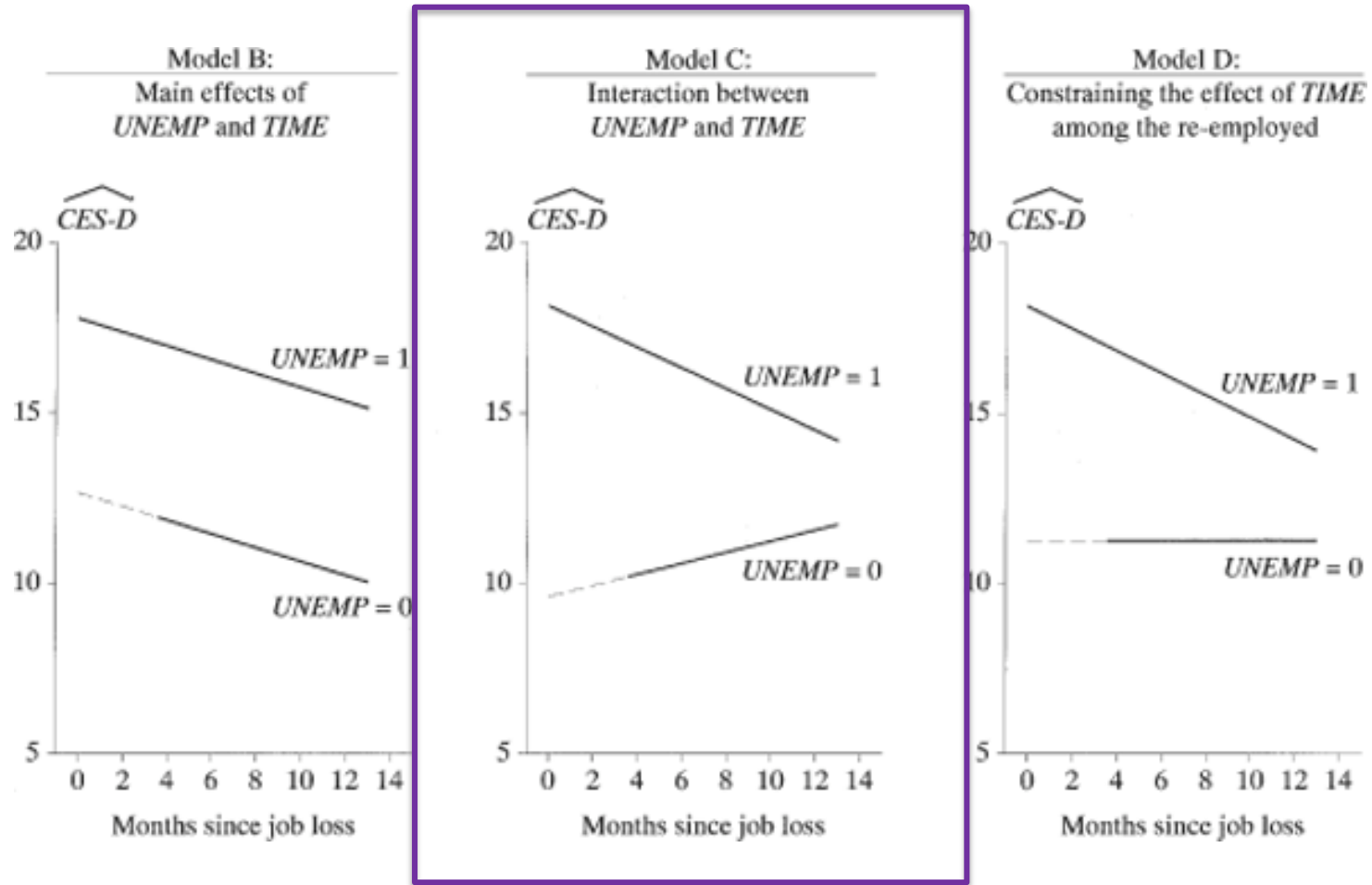
• But you need to think **very** carefully about the hypothesized error structure:

- *We've basically added another level-1 parameter to capture the interaction*
- *Just like we asked for the main effect of the TV predictor UNEMP, should we allow the interaction effect to vary across people?*
- *We won't right now, but we will in a minute.*

What happens when we fit Model C to data?

Time-varying predictor and the slope

(Singer and Willet pg. 167, Figure 5.4)



Time-varying predictor and the slope

(Singer and Willet pg. 163, Figure 5.7)

Model A

- Unconditional growth model

Model B

- The main effect model

Model C

- Addition of interaction term between time and unemployment

$$Y_{ij} = [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{20}UNEMP_{ij} + \gamma_{30}UNEMP_{ij} \times TIME_{ij}] + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Model D

- Includes fixed and random effects of unemployment

$$Y_{ij} = [\gamma_{00} + \gamma_{20}UNEMP_{ij} + \gamma_{30}UNEMP_{ij} \times TIME_{ij}] + [\zeta_{0i} + \zeta_{2i}UNEMP_{ij} + \zeta_{3i}UNEMP_{ij} \times TIME_{ij} + \varepsilon_{ij}]$$

- No improvement from model B
- May not be the best model

Table 5.7: Results of fitting a taxonomy of multilevel models for change to the unemployment data ($n = 254$)

		Parameter	Model A	Model B	Model C	Model D
Fixed Effects						
Composite	Intercept	γ_{00}	17.6694**	12.6656***	9.6167***	11.2666***
model	(initial status)		(0.7756)	(1.2421)	(1.8893)	(0.7690)
	<i>TIME</i>	γ_{10}	-0.4220***	-0.2020*	0.1620	
	(rate of change)		(0.0830)	(0.0933)	(0.1937)	
	<i>UNEMP</i>	γ_{20}		5.1113***	8.5291***	6.8795***
				(0.9888)	(1.8779)	(0.9133)
	<i>UNEMP</i> by <i>TIME</i>	γ_{30}			-0.4652*	-0.3254**
					(0.2172)	(0.1105)
Variance Components						
Level-1:	Within-person	σ^2_{ε}	68.85***	62.39***	62.03***	62.43***
Level-2:	In intercept	σ^2_0	86.85***	93.52***	93.71***	41.52***
	In rate of Change	σ^2_1	0.36*	0.46**	0.45**	—
	In <i>UNEMP</i>	σ^2_2	—	—	—	40.45*
	In <i>UNEMP</i> by <i>TIME</i>	σ^2_3	—	—	—	0.71**
Goodness-of-fit						
	Deviance		5133.1	5107.6	5103.0	5093.6
	AIC		5145.1	5121.6	5119.7	5113.6
	BIC		5166.3	5146.4	5147.3	5148.9

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Model C: Allowing the effect of a TV predictor to vary over time

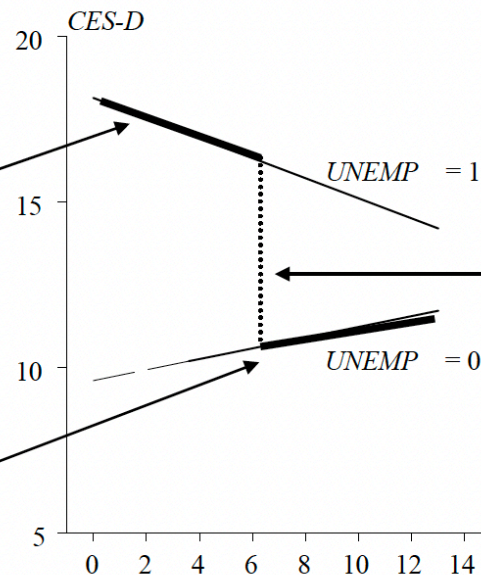
		Parameter	Model B	Model C
Fixed Effects				
Composite model	Intercept	γ_{00}	12.6656*** (1.2421)	9.6167*** (1.8893)
	TIME (rate of change)	γ_{10}	-0.2020* (0.0933)	0.1620 (0.1937)
	UNEMP	γ_{20}	5.1113*** (0.9888)	8.5291*** (1.8779)
	UNEMP by TIME	γ_{30}		-0.4652* (0.2172)
Variance Components				
Level-1:	Within-person	σ^2_{ϵ}	62.39***	62.03***
Level-2:	In intercept	σ^2_0	93.52***	93.71***
	In rate of change	σ^2_1	0.46**	0.45**
Goodness-of-fit				
	Deviance		5107.6	5103.0
	AIC		5121.6	5119.7
	BIC		5146.4	5147.3

- $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Main effect of TIME is now positive (!)
& not stat sig ?!?!?!?!?!?!?!?!?

*UNEMP*TIME interaction is stat sig*
($p < .05$)

Model B is a much poorer fit than C
(Δ Deviance = 4.6, 1 df, $p < .05$)



*What about people
who get a job?*

Should the trajectory for the reemployed be constrained to 0?

Consistently unemployed (UNEMP=1)

$$\begin{aligned}\hat{Y}_j &= (9.6167 + 8.5291) \\ &\quad + 0.(0.1620 - 0.4652)MONTHS_j \\ \hat{Y}_i &= 18.1458 - 0.3032MONTHS_i\end{aligned}$$

Consistently employed (UNEMP=0)

$$\hat{Y}_j = 9.6167 + 0.1620 MONTHS_j$$

Time-varying predictor and the slope

(Singer and Willet pg. 163, Figure 5.7)

Upon layoff, the average unemployed person in the population has a CES-D score of 18.15 (= 11.27 + 6.88)

Over time, as he/she acclimates to being unemployed, the average unemployed person's CES-D scores decline at a rate of -0.33 per month ($p < .01$)

Table 5.7: Results of fitting a taxonomy of multilevel models for change to the unemployment data ($n = 254$)

		Parameter	Model A	Model B	Model C	Model D
Fixed Effects						
Composite model	Intercept	Y_{00}	17.6694**	12.6656***	9.6167***	11.2666***
	(initial status)		(0.7756)	(1.2421)	(1.8893)	(0.7690)
	<i>TIME</i>	Y_{10}	-0.4220***	-0.2020*	0.1620	
	(rate of change)		(0.0830)	(0.0933)	(0.1937)	
	<i>UNEMP</i>	Y_{20}		5.1113***	8.5291***	6.8795***
				(0.9888)	(1.8779)	(0.9133)
	<i>UNEMP</i> by <i>TIME</i>	Y_{30}			-0.4652*	-0.3254**
					(0.2172)	(0.1105)
Variance Components						
Level-1:	Within-person	σ^2_{ϵ}	68.85***	62.39***	62.03***	62.43***
Level-2:	In intercept	σ^2_0	86.85***	93.52***	93.71***	41.52***
	In rate of Change	σ^2_1	0.36*	0.46**	0.45**	—
	In <i>UNEMP</i>	σ^2_2	—	—	—	40.45*
	In <i>UNEMP</i> by <i>TIME</i>	σ^2_3	—	—	—	0.71**
Goodness-of-fit						
	Deviance		5133.1	5107.6	5103.0	5093.6
	AIC		5145.1	5121.6	5119.7	5113.6
	BIC		5166.3	5146.4	5147.3	5148.9

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Model significantly better than Model C

Model D: Constraining the individual growth trajectory among the reemployed

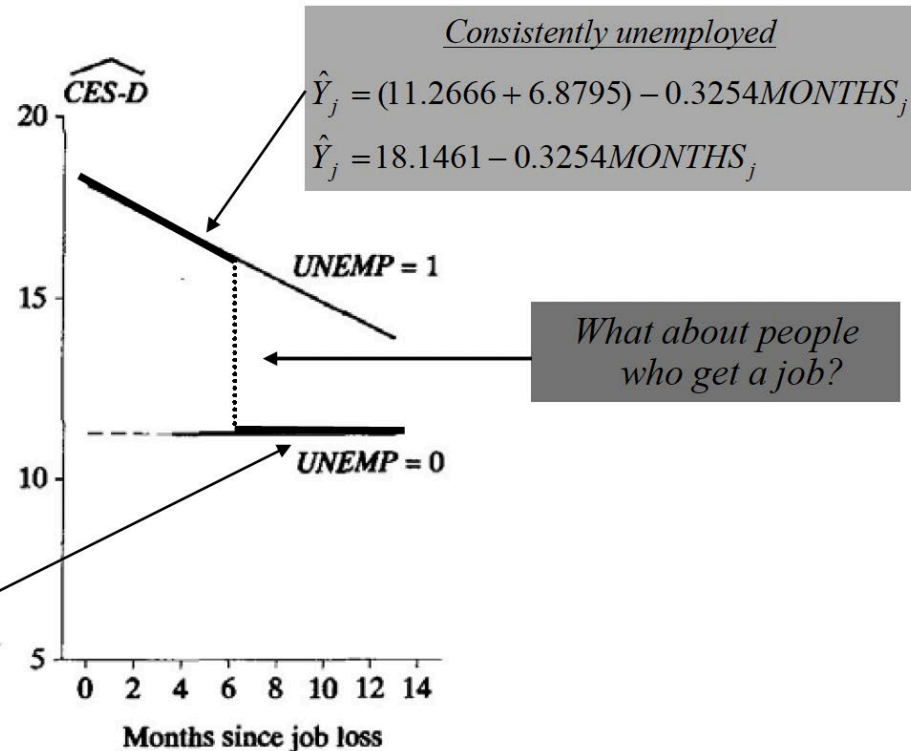
		Parameter	Model C	Model D
Fixed Effects				
Composite model	Intercept (initial status)	γ_{00}	9.6167*** (1.8893)	11.2666*** (0.7690)
	TIME (rate of change)	γ_{10}	0.1620 (0.1937)	
	UNEMP	γ_{20}	8.5291*** (1.8779)	6.8795*** (0.9133)
	UNEMP by TIME	γ_{30}	-0.4652* (0.2172)	-0.3254** (0.1105)
Variance Components				
Level-1:	Within-person	σ_e^2	62.03***	62.43***
Level-2:	In intercept	σ_0^2	93.71***	41.52***
	In rate of change	σ_1^2	0.45**	—
	In UNEMP	σ_2^2	—	40.45*
	In UNEMP by TIME	σ_3^2	—	0.71**
Goodness-of-fit				
	Deviance		5103.0	5093.6
	AIC		5119.7	5115.6
	BIC		5147.3	5148.9

— $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Best fitting model (lowest AIC and BIC)

Consistently employed

$$\hat{Y}_j = 11.2666$$



Back to Variance for a Moment

Time-varying predictor and the slope

(Singer and Willet pg. 170-171, Figure 5.4)

Variance Components

Time invariant-covariates

- σ_{ϵ}^2 stays relatively stable because time-invariant predictors cannot explain within-person variance
- σ_0^2 and σ_1^2 Would decline if time invariant covariate explained some of the between person variance for the intercept and slope, respectively

Time varying-covariates

1. All three variance components can change because they vary both between- and within-person
2. Can usually interpret decrease in within-person variance component, but not between-person variance component

Time-varying predictor and the slope

(Singer and Willet pg. 170-171, Figure 5.4)

Table 5.7: Results of fitting a taxonomy of multilevel models for change to the unemployment data ($n = 254$)

		Parameter	Model A	Model B	Model C	Model D
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model	(initial status)		(0.7756)	(1.2421)	(1.8893)	(0.7690)
	<i>TIME</i>	γ_{10}	-0.4220***	-0.2020*	0.1620	
	(rate of change)		(0.0830)	(0.0933)	(0.1937)	
	<i>UNEMP</i>	γ_{20}		5.1113***	8.5291***	6.8795***
				(0.9888)	(1.8779)	(0.9133)
	<i>UNEMP</i> by	γ_{30}			-0.4652*	-0.3254**
	<i>TIME</i>				(0.2172)	(0.1105)
Variance Components						
Level-1:	Within-person	σ^2_{ϵ}	68.85***	62.39***	62.03***	62.43***
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Goodness-of-fit						
	Deviance		5133.1	5107.6	5103.0	5093.6
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~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Variance Components

9.4% reduction meaning that unemployment status explains about 9% of the variance in CES-D (see 4.4.3)

These two parameters should not be compared. Instead focus on the within-person variance component and the changing fit statistics

All three fit indices improve from model A to model B



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Race as a Social Determinant of Sleep Health from 2017-2022: Exploring the Intersection of Race and Gender

Stephanie H. Cook, DrPH, MPH, Antoneta Karaj, MS, & Mariana Rodrigues, MA



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- Short sleep duration, defined as less than 7 hours of sleep in a 24 hours period, is a key indicator of poor sleep health
- Lack of minutes of sleep is associated with cardiovascular disease
- Black and Hispanics report less minutes of sleep per night compared to whites
- Findings concerning differences between biological males and females mixed
- Women with poorer sleep health may be more likely to die earlier than men



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1. Within the category of race/ethnicity, we hypothesized that racial/ethnic minority women would have less minutes of sleep than white women
2. Within the category of race/ethnicity, we hypothesized that racial/ethnic minority men would have less minutes of sleep than white men
3. Non-white women would have decreasing trajectories of sleep duration (fewer minutes of sleep) compared to white women.



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- Minutes asleep: Measured by participants' Fitbit data. Minutes asleep derived from the daily summaries recorded on a participants' Fitbit for those participants who wore the device for at least 10 hours in a 24 hour period.
- Race/Ethnicity: Measured via self-report after enrollment via an online questionnaire.
- Gender: Measured via self-report after enrollment via an online questionnaire.
- Covariates (BMI, Heart conditions, substance use): Measured via self-report after enrollment via an online questionnaire.



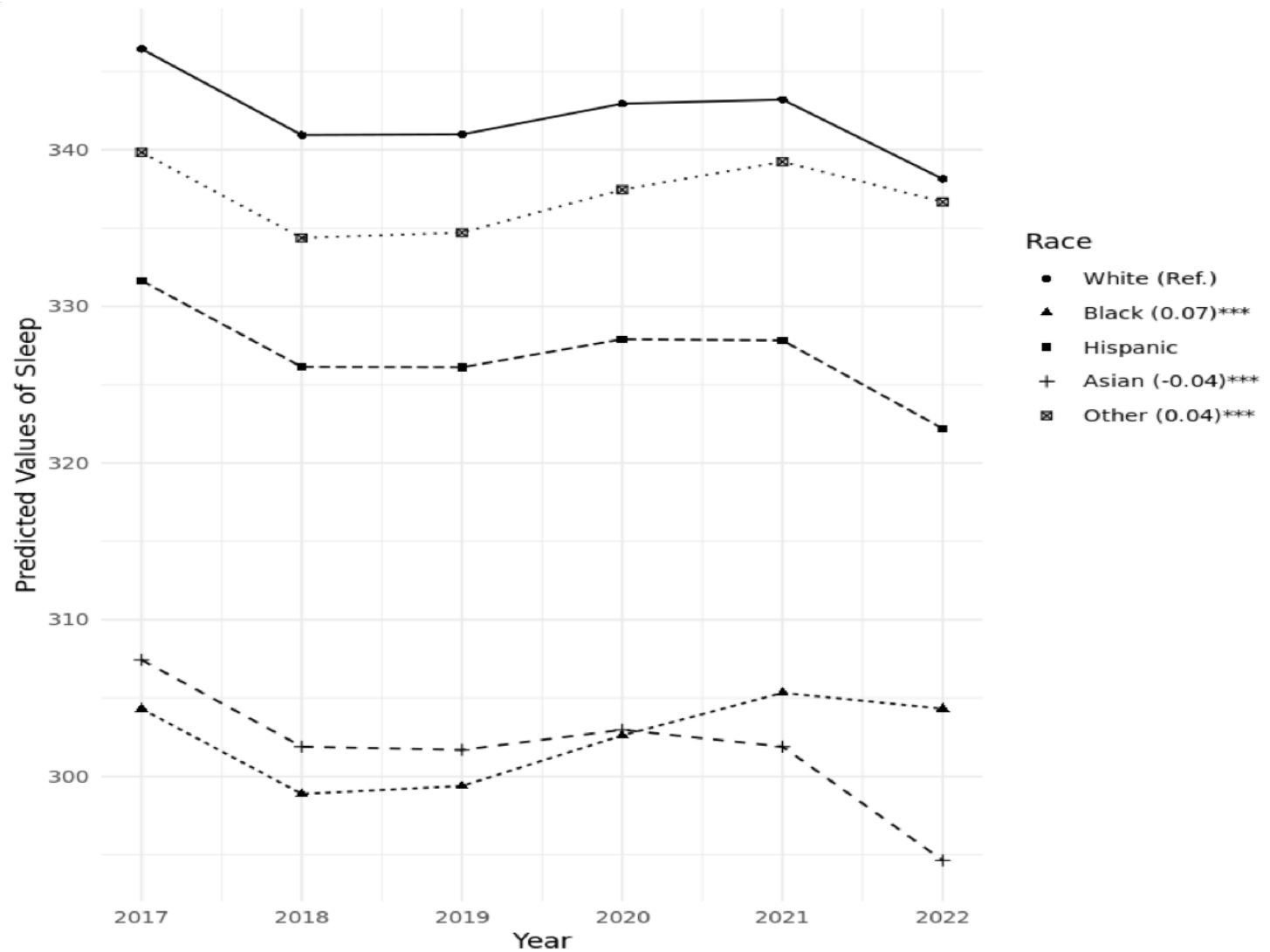
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- Bivariate analyses were conducted to observe potential associations between the main variables of interest.
- Unconditional and conditional linear mixed models were conducted to address the study hypotheses. To test the intersectionality framework, an interaction term between race and gender was used (Guan et al., 2021)
- All analyses conducted in R in the *All of Us* Researcher Workbench.



Minutes Asleep Trajectories by Race

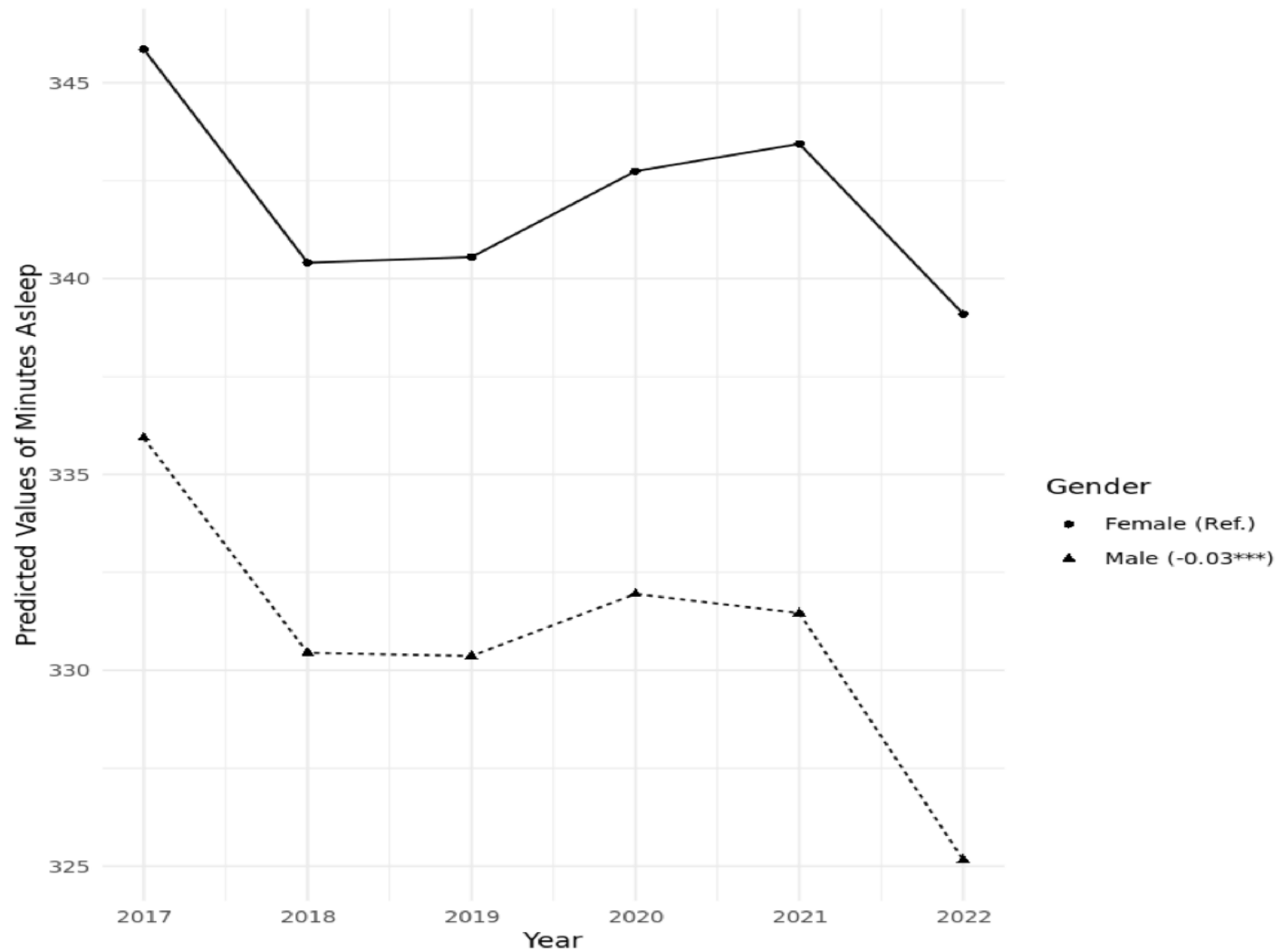




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Minutes Asleep Trajectories by Gender

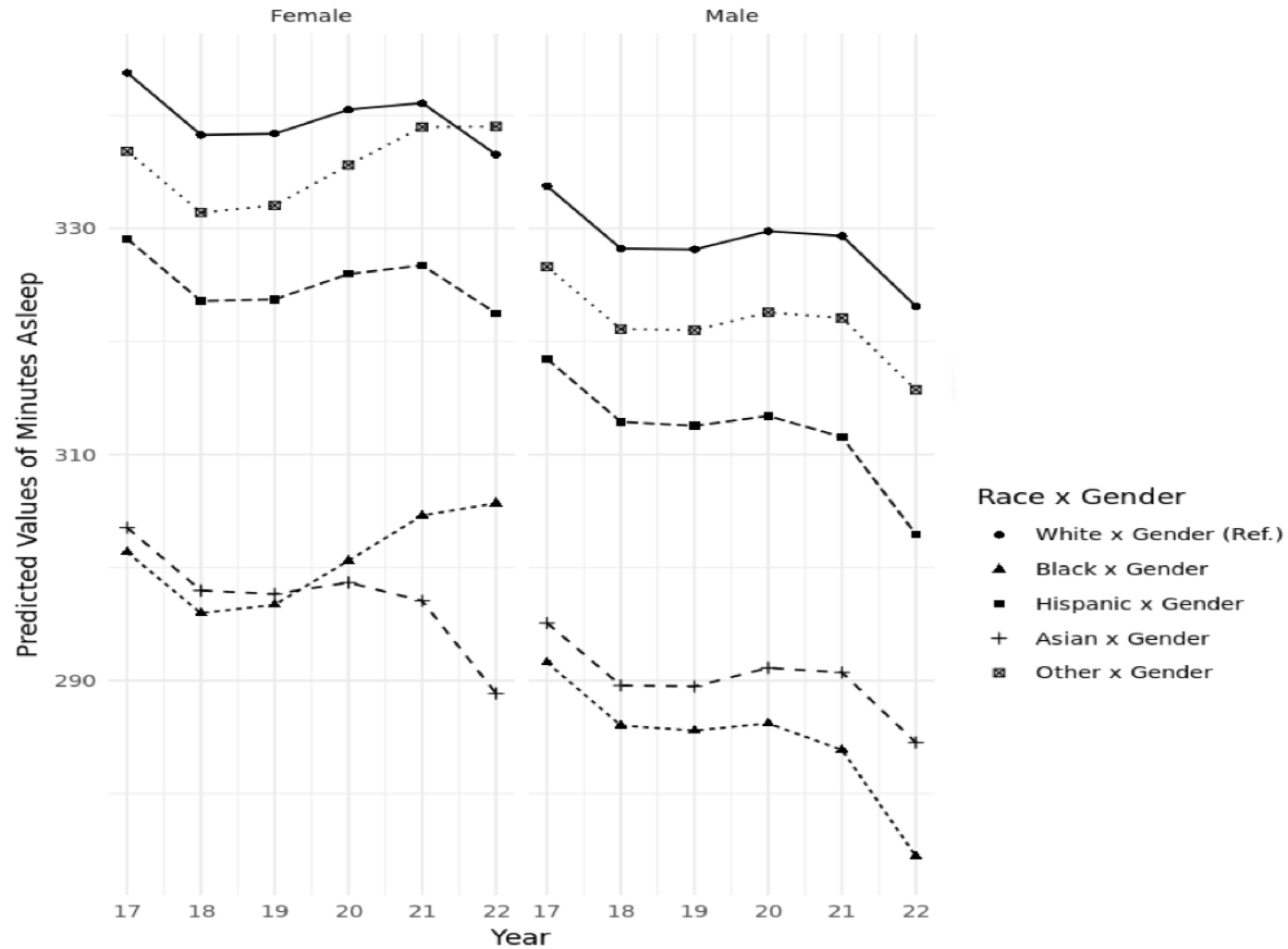




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Minutes Asleep Trajectories by Race x Gender





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- Our hypotheses were partially confirmed.
- Among racial/ethnic minorities, males had less minutes of sleep compared to females, which contradicts our original hypothesis.
- Within the category of gender racial/ethnic minority females had less minutes of sleep overall than white females and the same was true within the male category.
- Future SDoH research related to sleep should take an intersectionality approach in order to uncover invisible processes. In doing so, prevention interventions concerning sleep can be better tailored to the needs of specific populations.

Next Class...

- **More on time** (Singer and Willet pgs. 139-159)
 - Variably spaced measurement occasions
 - Variable measurement occasions
 - Missing data concerns
 - Fixes for unbalanced data
- **Discontinuous individual change** (Singer and Willet pgs. 189-208)